ModelIt!

Student Activity Pack

Rock Cycle Deep Dive: From Magma to Mountain

How Earth Recycles Itself Over Millions of Years

Grade 07L2 | NGSS MS-ESS2-1, MS-ESS1-4 | Energy and Matter

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Where do you think rocks come from? Are they all the same?

2 What do you think happens to a rock if it gets buried really deep underground for millions of years?

3 Draw what you think happens when a volcano erupts and lava flows onto the surface.

4 How do you think scientists figure out which rocks are older and which are younger?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Heat & Pressure (Depth) | Weathering Rate | Igneous Rock Formation | Sedimentary Layers | Metamorphic Transformation

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: High Heat & Pressure

Intense conditions deep underground, minimal surface erosion

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: High Weathering

Rapid surface erosion, minimal underground heat

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Balanced Conditions

Moderate heat/pressure AND moderate weathering

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Tectonic Plate Movement — The motion of Earth's crustal plates that subducts rock into the mantle and uplifts it to the surface

Erosion Transport Distance — How far weathered sediment travels from its source before being deposited and compacted

Fossil Preservation — The likelihood that organisms become fossilized during sedimentary rock formation, dependent on burial speed and conditions

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

If Earth is 4.6 billion years old, why aren't all rocks ancient? How does Earth constantly create, destroy, and recreate rocks? Use evidence from your simulation to support your answers.

1. **Why is the rock cycle called a 'cycle' rather than a 'chain'? How does your model demonstrate this?**

2. **How does the rock cycle explain why we find marine fossils on mountain tops?**

3. **What would happen to the rock cycle if Earth's internal heat completely died out?**

4. **Why are sedimentary rocks the most important for understanding Earth's biological history?**

5. **How does your model show that the rock cycle is driven by BOTH internal and external energy sources?**

8 STEM Challenge: Build a Geologic History Timeline

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YOUR ENGINEERING MISSION

Create a geologic history timeline showing how the rock cycle has shaped Earth's surface over billions of years, using rock strata evidence and your simulation model data.

THE SCENARIO

A geology museum is developing a new exhibit on Earth's 4.6-billion-year history. They need your team to create a visual timeline showing how the rock cycle has continuously reshaped the planet, using rock strata as evidence for major geologic events.

Engineering Design Thinking

1. WHAT MAJOR GEOLOGIC EVENTS WILL YOU INCLUDE IN YOUR TIMELINE?
2. HOW WILL YOU SHOW THE RELATIONSHIP BETWEEN THE ROCK CYCLE AND THE GEOLOGIC TIME SCALE?
3. WHAT MODEL DATA WILL YOU USE TO EXPLAIN HOW DIFFERENT CONDITIONS CREATE DIFFERENT ROCKS?
4. HOW WILL YOU MAKE 4.6 BILLION YEARS OF HISTORY ACCESSIBLE AND ENGAGING FOR MUSEUM VISITORS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Geologists study Earth's materials, processes, and history. They work in natural resource exploration, environmental consulting, hazard assessment, and academic research, earning $65,000-$120,000/year.

9 Post-Assessment

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Show What You Learned!

Multiple Choice: Now that you have completed the lesson, circle the best answer for each question. Compare your answers to your pre-assessment — did your thinking change?

1 A geologist finds a rock that was originally sandstone (sedimentary) but now has a crystalline, folded structure. Based on the rock cycle model, which process and resulting rock type best explains this transformation?

- **A.** Weathering transformed the sandstone into igneous rock at the surface
- **B.** The sandstone was buried deep enough to be subjected to intense heat and pressure, transforming it into metamorphic rock without melting
- **C.** The sandstone melted completely and resolidified as a new sedimentary rock
- **D.** Chemical reactions with rainwater dissolved and reformed the sandstone into metamorphic rock at the surface

2 In the rock cycle model, Sedimentary Layers have a positive relationship with Metamorphic Transformation. Which mechanism explains why the accumulation of sedimentary layers can lead to metamorphism?

- **A.** Sedimentary rocks contain chemicals that trigger metamorphic reactions on their own
- **B.** As sedimentary layers accumulate, the weight of upper layers creates increasing pressure on deeper layers. Combined with geothermal heat at depth, this triggers metamorphic transformation
- **C.** Sedimentary rocks automatically become metamorphic after a certain age regardless of depth
- **D.** Metamorphic transformation occurs at the surface where sedimentary layers are deposited

3 The model shows that the rock cycle has no beginning or end, meaning any rock type can become any other rock type. A student argues that sedimentary rock must always become metamorphic before becoming igneous. Using the model, which scenario disproves this claim?

- **A.** Sedimentary rock can be subducted directly into the mantle where extreme heat melts it into magma, which cools into igneous rock, bypassing the metamorphic stage entirely
- **B.** Sedimentary rock always follows the sequence: sedimentary to metamorphic to igneous to sedimentary
- **C.** The rock cycle only operates in one direction, so the student's claim is correct
- **D.** Sedimentary rock cannot become igneous rock under any circumstances

4 In the model, when Heat & Pressure is set to low and Weathering Rate is set to high, Sedimentary Layers dominate while Igneous Rock Formation and Metamorphic Transformation are minimal. If Earth's internal heat source completely shut off, what would the model predict for the long-term rock cycle?

- **A.** All three rock types would continue forming at equal rates because the Sun provides enough energy for all processes
- **B.** Only weathering and sedimentary formation would continue because the Sun drives surface erosion, while metamorphic and igneous processes would stop without internal heat
- **C.** The rock cycle would immediately stop completely because both energy sources are needed for any process
- **D.** Igneous rock formation would increase because surface temperatures from the Sun would melt rocks

5 **Geologists use the Law of Superposition to determine relative rock ages: in undisturbed strata, older layers are on the bottom and younger layers are on top. A scientist discovers a fossil of a marine organism embedded in rock near a mountain summit. Using the rock cycle model, which sequence of events best explains this finding?**

A. The marine organism climbed to the mountain top before becoming fossilized

B. Marine sediment containing the organism was deposited on the ocean floor, lithified into sedimentary rock, then tectonic forces uplifted the rock from the seafloor to mountain height over millions of years

C. Volcanic eruptions carried marine fossils from the ocean floor and deposited them on mountaintops

D. Rain washed marine organisms uphill during a massive flood